EXPERIMENT NO. 1 IDENTIFICATION OF INFORMATION EXPOSURE RISKS THROUGH SOCIAL MEDIA AND INTERNET SOURCES

AIM:

To understand Human Security Assessment by identifying how sensitive information can be exposed through social media, public websites, internet sources, and human interactions. It also helps evaluate phishing awareness, trust verification, and security practices using passive reconnaissance, network diagnostics, and email authentication analysis.

OBJECTIVES:

1. To understand Human Security Assessment and its role in preventing cyber threats.
2. To study social engineering and identify information exposed through public sources.
3. To learn basic network diagnostic commands for analyzing network configuration and connectivity.
4. To understand email authentication (SPF, DKIM, and DMARC) for phishing prevention.
5. To identify security risks and recommend measures to improve organizational security.

HARDWARE & SOFTWARE REQUIREMENTS

Category	Specification & Tools
Hardware Configuration	PC/Laptop with Intel Core i5 or higher, minimum 4 GB RAM, Internet Connection
Operating System	Windows 10/11
Network Diagnostic Utilities	ipconfig, ping, tracert, nslookup, netstat
Analysis & Audit Tools	WHOIS, Nmap, PhishTool, OSINT Framework, Google Chrome
Documentation Tools	Microsoft Word

These tools are used for information gathering, network diagnostics, and Human Security Assessment.

THEORY:

Human Security Assessment is an important part of cybersecurity that focuses on identifying security risks caused by people, public information, and organizational practices. Instead of only finding software or hardware vulnerabilities, it examines how attackers can collect publicly available information and use it for phishing, social engineering, or identity impersonation.

Information shared on social media, websites, and other public sources can reveal details about an organization, its employees, and its network. Attackers use this information to create convincing phishing emails or other social engineering attacks. Human Security Assessment helps identify these risks before they are exploited and improves an organization's overall security through awareness and preventive measures.

HUMAN SECURITY ASSESSMENT - PHASES, TOOLS & COMMANDS:

1. **Visibility Audit (Passive Reconnaissance)**
   In this phase, publicly available information is collected from websites, search engines, and social media without directly interacting with the target. The objective is to identify exposed details such as email addresses, contact numbers, staff information, and organizational structure that could be misused by attackers.

Tool / Utility	Purpose	Command / Method Used
Web Browser	Access public websites and social media	Website browsing
WHOIS	Retrieve domain registration details	whois <domain_name>
OSINT Framework	Collect publicly available information	Web-based OSINT resources

2. **Active Detection & Device Intelligence**
   This phase uses basic Windows networking commands such as ipconfig, ping, tracert, nslookup, and netstat to examine network configuration, verify connectivity, identify IP addresses, and view active network connections.

Tool / Utility	Purpose	Command Used
IPCONFIG	Display local network configuration	ipconfig
PING	Check network connectivity	ping <hostname>

Tool / Utility	Purpose	Command Used
TRACERT	Trace the route to a destination	tracert <hostname>
NSLOOKUP	Retrieve DNS records and IP address	nslookup <domain_name>
NETSTAT	Display active network connections	netstat -an
Nmap	Scan hosts and open ports	nmap <IP Address>

3. **Trust Verification & Engagement**
   This phase focuses on email security by examining email headers and verifying SPF, DKIM, and DMARC authentication records to detect phishing attempts and email spoofing.

Tool / Utility	Purpose	Method Used
PhishTool	Analyze email headers and verify SPF, DKIM, and DMARC	Email Header Analysis

IMPLEMENTATION STEPS:

1. **Visibility Audit (Passive Reconnaissance)**
   - Selected openai.com as the target domain for passive reconnaissance.
   - Accessed the OSINT Framework and used the WHOIS Records tool to gather domain registration information.
   - Performed a WHOIS lookup on openai.com to obtain registrar, creation date, expiry date, and name server details.
   - Used nslookup openai.com to retrieve DNS records and the IP address.
   - Recorded and analyzed the collected information for security assessment.
2. **Active Detection & Device Intelligence**
   - Opened Command Prompt in Windows.
   - Executed the ipconfig command to view the local network configuration.
   - Used the ping command to test network connectivity.
   - Executed the tracert command to trace the route to the target host.
   - Used netstat to view active network connections.
   - Performed a basic Nmap scan to identify open ports and active services.

3. **Trust Verification & Engagement**
    - Obtained a sample email header for analysis.
    - Analyzed the email header using PhishTool.
    - Verified the SPF, DKIM, and DMARC authentication results.

OBSERVATIONS:

During the Human Security Assessment practical, publicly available information, network details, and email authentication records were analyzed to identify information exposure risks and understand basic security assessment techniques.

- Publicly available domain information, including registration details, name servers, and DNS records, was successfully collected using OSINT Framework, WHOIS, and NSLOOKUP.
- Network diagnostic tools and Nmap were used to examine network configuration, verify connectivity, and identify active network services and open ports.
- PhishTool analyzed the email header and verified SPF, DKIM, and DMARC authentication to assess the authenticity of the email and detect potential spoofing.

OUTPUT:

Phase 1: Visibility Audit

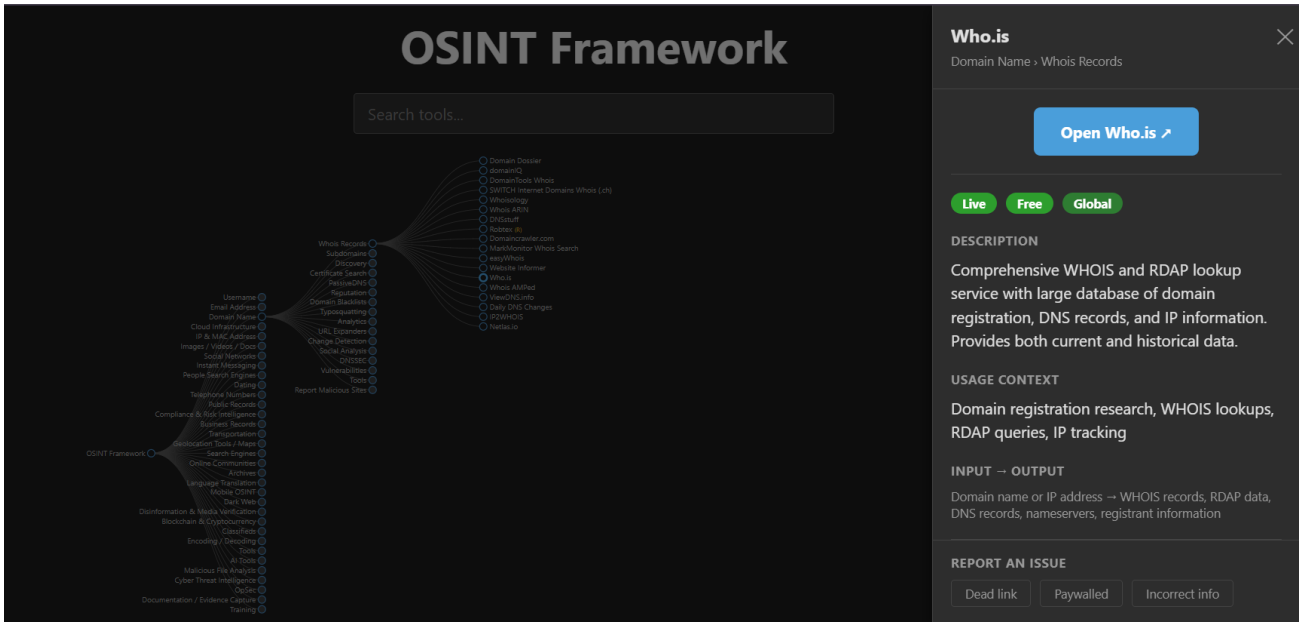


Fig. 1: Accessing WHOIS via OSINT Framework

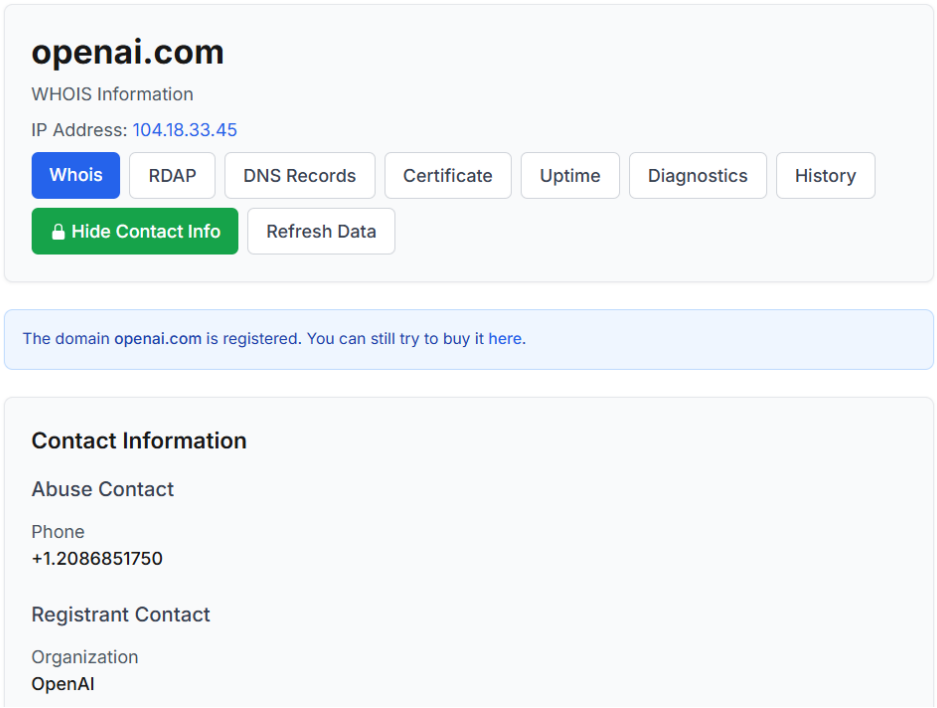


Fig. 2: WHOIS lookup results for the target openai.com

```
C:\Users\salga\Downloads\WhoIs>whois openai.com

Whois v1.21 - Domain information lookup
Copyright (C) 2005-2019 Mark Russinovich
Sysinternals - www.sysinternals.com

Connecting to COM.whois-servers.net...

WHOIS Server: whois.markmonitor.com
   Registrar URL: http://www.markmonitor.com
   Updated Date: 2024-10-17T22:20:04Z
   Creation Date: 2007-01-19T19:28:24Z
   Registry Expiry Date: 2029-01-19T19:28:24Z
   Registrar: MarkMonitor Inc.
   Registrar IANA ID: 292
   Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
   Registrar Abuse Contact Phone: +1.2086851750
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
   Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
   Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
   Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
   Name Server: NS1-02.AZURE-DNS.COM
   Name Server: NS2-02.AZURE-DNS.NET
   Name Server: NS3-02.AZURE-DNS.ORG
   Name Server: NS4-02.AZURE-DNS.INFO
   DNSSEC: unsigned
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-29T12:09:28Z <<<
```

Fig. 3: WHOIS Information using Command Prompt

```
C:\Users\salga>nslookup openai.com
Server:  jiofiber.local.html
Address:  192.168.31.1

Non-authoritative answer:
Name:    openai.com
Addresses:  64:ff9b::6812:212d
          64:ff9b::ac40:9ad3
          104.18.33.45
          172.64.154.211
```

Fig. 4: DNS Information using NSLOOKUP

Phase 2: Active Detection & Device Intelligence

```
C:\Users\salga>ping openai.com

Pinging openai.com [172.64.154.211] with 32 bytes of data:
Reply from 172.64.154.211: bytes=32 time=13ms TTL=51
Reply from 172.64.154.211: bytes=32 time=43ms TTL=51
Reply from 172.64.154.211: bytes=32 time=57ms TTL=51
Reply from 172.64.154.211: bytes=32 time=54ms TTL=51

Ping statistics for 172.64.154.211:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 13ms, Maximum = 57ms, Average = 41ms
```

Fig. 5: Network Connectivity Test using PING

```
C:\Users\salga>tracert openai.com

Tracing route to openai.com [172.64.154.211]
over a maximum of 30 hops:

  1    12 ms     3 ms     1 ms  router [192.168.31.1]
  2     3 ms     3 ms     3 ms  192.0.0.1
  3     *        *        *     Request timed out.
  4     9 ms     9 ms    10 ms  192.0.0.1
  5    58 ms    12 ms     9 ms  192.0.0.1
  6    22 ms    10 ms     8 ms  192.0.0.1
  7    24 ms    10 ms     9 ms  172.26.76.194
  8    52 ms    16 ms    38 ms  192.168.53.176
  9     *        *        *     Request timed out.
 10     *        *        *     Request timed out.
 11     *        *        *     Request timed out.
 12    25 ms    22 ms    20 ms  162.158.226.83
 13    27 ms     9 ms     9 ms  172.64.154.211

Trace complete.
```

Fig. 6: Route Tracing using TRACERT

```
C:\Users\salga>ipconfig

Windows IP Configuration


Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::ee98:b3aa:c51e:b7d5%45
   IPv4 Address. . . . . . . . . . . : 172.26.16.1
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : lan
   IPv6 Address. . . . . . . . . . . : 2409:4090:1f:9e02:5792:837f:92ce:485a
   Temporary IPv6 Address. . . . . . : 2409:4090:1f:9e02:c8bc:dbe1:3777:662c
   Link-local IPv6 Address . . . . . : fe80::1731:fdd5:359a:1934%14
   IPv4 Address. . . . . . . . . . . : 192.168.31.185
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::be9d:4eff:fed4:853%14
                                       192.168.31.1
```

Fig. 7: Network Configuration using IPCONFIG

```
C:\Users\salga>netstat -an

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:5432           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49668          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49669          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49672          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49680          0.0.0.0:0              LISTENING
  TCP    127.0.0.1:49712        127.0.0.1:49713        ESTABLISHED
  TCP    127.0.0.1:49713        127.0.0.1:49712        ESTABLISHED
  TCP    127.0.0.1:49715        127.0.0.1:49716        ESTABLISHED
  TCP    127.0.0.1:49716        127.0.0.1:49715        ESTABLISHED
  TCP    127.0.0.1:49717        127.0.0.1:49718        ESTABLISHED
  TCP    127.0.0.1:49718        127.0.0.1:49717        ESTABLISHED
  TCP    172.26.16.1:139        0.0.0.0:0              LISTENING
  TCP    192.168.31.185:139     0.0.0.0:0              LISTENING
  TCP    192.168.31.185:9084    192.178.173.94:443     ESTABLISHED
  TCP    192.168.31.185:20450   192.168.31.219:8009    ESTABLISHED
```

Fig. 8: Active Network Connections using NETSTAT

```
C:\Users\salga>nmap 104.18.33.45
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-29 19:07 +0530
Nmap scan report for 104.18.33.45
Host is up (0.013s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT     STATE SERVICE
80/tcp   open  http
443/tcp  open  https
8080/tcp open  http-proxy
8443/tcp open  https-alt

Nmap done: 1 IP address (1 host up) scanned in 6.01 seconds
```

Fig. 9: Port Scanning using Nmap

Phase 3: Trust Verification & Engagement using PhishTool

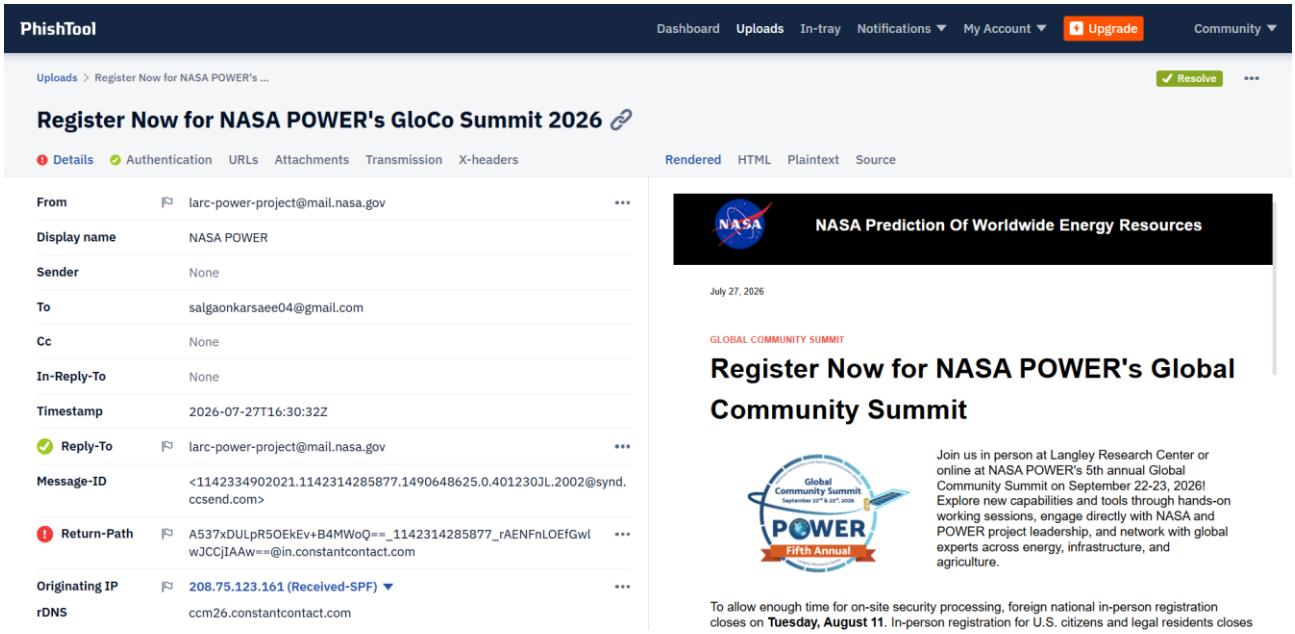


Fig. 10: Email Analysis Report in PhishTool

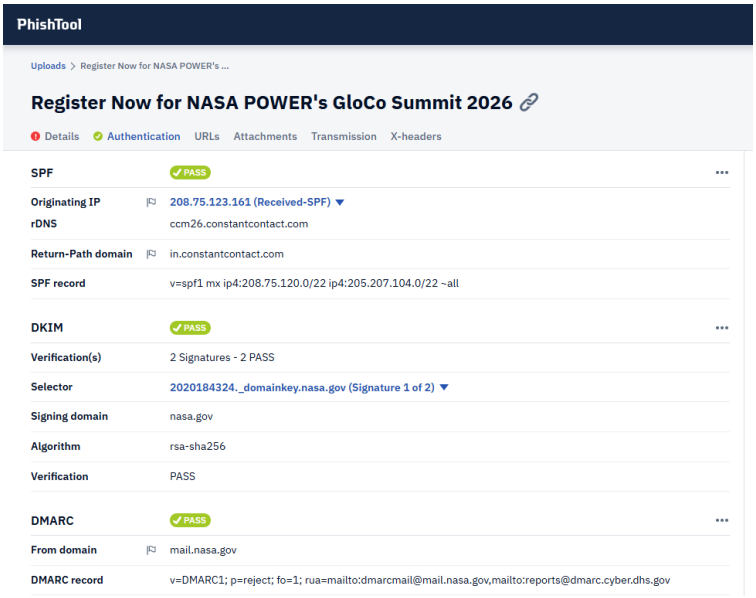


Fig. 11: Email Authentication Results (SPF, DKIM, and DMARC)

ANALYSIS:

The practical demonstrated how publicly available information, network details, and email analysis can reveal potential security risks. The collected information helped understand how attackers perform reconnaissance and how security measures such as proper network configuration and email authentication can reduce these risks.

Overall, the experiment highlighted the importance of protecting sensitive information and improving security awareness to prevent cyber threats.

PREVENTIVE MEASURES:

1. Avoid sharing sensitive organizational information on public platforms and websites.
2. Conduct regular cybersecurity awareness training to reduce phishing and social engineering risks.
3. Implement SPF, DKIM, and DMARC to improve email security and prevent spoofing.
4. Regularly monitor network services, open ports, and exposed information to reduce security risks.
5. Keep systems updated, use strong passwords, and enable security controls such as Multi-Factor Authentication (MFA).

CONCLUSION:

The practical provided an understanding of how information exposure, network analysis, and email security can affect an organization's security. The tools used helped identify possible risks and the importance of following proper security practices**.**